

3.1 The Power Supply Unit

The role of the Power Supply Unit (PSU) is easily overlooked—since it comes bundled with the cabinet—and rarely gets special mention. With more and more power-hungry components being plugged into PCs, the need for a proper PSU becomes important.

3.1.1 Amperage

This refers to the value of the current carried by a power cable, and is expressed in Amperes. Each of the voltage “rails” has an associated amperage. Since 12 V is used to drive all the motors (hard disk, optical drives), and also power the graphics card(s) (in the case of additional power being needed), it is recommended to buy a power supply unit with higher amperage in the 12 V rail.

3.1.2 ATX Power Supply Unit

PSUs are available in different form factors, with the ATX form factor being the most widely-used. The ATX power supply is not directly controlled by the user; the motherboard controls it. ATX specifications have been updated from the original so as to support additional, power-hungry components. The latest version is 2.2—it includes specifications for the 24-pin motherboard power connector (the original specifications included only a 20-pin power connector), the 12 V 4-pin motherboard power connector, and the 6-pin PCIe graphics card power connector.

3.1.4 Efficiency

Energy is lost during the conversion from AC to DC—mostly in the form of heat. The greater the amount of DC generated given a certain AC input, the better the efficiency of the unit. The quality of components used in the PSU influences its efficiency.

3.1.5 Modular Power Supply

These are power supplies that allow users to connect required cables only when needed, thus avoiding the clutter seen in most power supplies. These only have sockets for the required connections, though some have the motherboard connector preinstalled.